The Cost of Reliable Near-Optimal Set Maintenance with Arriving Arms

Deqi Zheng* Xiaoyang Xu*
Qiuzhen College, Tsinghua University

Yuhong Yang†
Yau Mathematical Sciences Center, Tsinghua University

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Abstract

We study reliable maintenance of all relatively near-optimal arms while collecting rewards when new arms arrive exogenously and remain available. Reward laws are stationary, but arrivals change the membership boundary. Reporting progress is measured by cumulative unresolved arm time outside a prescribed boundary band, with simultaneous correctness of all labels. One uncertain arrival can obstruct many earlier certifications. A common-reference lower bound quantifies this effect, and matching worst-schedule area bounds yield an arrival-growth boundary for joint sublinear performance. On specified repeated-update families, temporary membership tasks impose rapid-maintenance regret costs, while separate constructions attain deferred and intermediate service levels. We develop Flexible Certificate-Cost Planning (F-CCP) as a tunable implementation. It combines sequential evidence and local certificate planning with a chosen reward kernel and separate observable budgets for reward and certification. Its guarantees include label reliability, unconditional expected-regret protection, and service-rate area bounds that retain budget rejections. Fixed-checkpoint and activation-budgeted constructions remain distinct analytical witnesses. Frozen paired experiments show lower mean costs in many finite-budget comparisons, together with exceptions, heavier regret tails, and nonzero reporting failures. These results do not characterize a general regret–maintenance frontier.

Keywords: sequential experimentation; arriving arms; near-optimal set maintenance; dynamic regret; confidence sequences.

1 Introduction

Sequential experiments often acquire new candidates while earlier ones remain available. An allocation rule must learn which candidates yield high rewards. An experimenter may also need to know which of all available candidates are within a specified tolerance of the current best. These

*The first two authors contributed equally.

†Corresponding author: yyangsc@mail.tsinghua.edu.cn.

objectives need not favor the same observations. A candidate can be sufficiently inferior to receive little reward-driven exploration and still be difficult to classify relative to the near-optimality boundary.

We study these objectives together under permanent exogenous arrivals and stationary reward laws. If $r^*(t)$ is the largest mean among the arms available at time t, an arm is relatively near-optimal when its mean is at least $r^*(t) - \varepsilon$. Reports certify membership, certify exclusion, or leave the arm unresolved. We require one simultaneous correctness event for all labels at all times. To rule out indefinite abstention, we charge the cumulative unresolved time of arms whose means lie more than η from the membership boundary. The question is how much regret is necessary, and how much is attainable, under this reporting requirement.

New arrivals affect the task in two ways. An uncertain new candidate may be better than the incumbent by enough to invalidate many existing positive reports. Evidence about that one candidate is then relevant to many reports at once. A later improvement can also make an earlier difficult membership question unnecessary: a candidate close to the old boundary may be far below the new one. Its existing upper confidence bound can be combined with a new reference's lower bound to certify exclusion. Neither effect requires the reward law of an old arm to change.

Our observation model and regret comparator have established precedents. Permanent arrivals form a monotone special case of sleeping-bandit availability (Kleinberg et al., 2010); ballooning bandits already compare reward with the best arm currently available (Ghalme et al., 2021). Exogenous arrivals differ from drawing arms on demand from a reservoir (Berry et al., 1997) and from the disappearing actions of mortal bandits (Chakrabarti et al., 2008). Benchmark improvements here arise from new arms, whereas nonstationary bandits allow the reward distributions of existing arms to change (Besbes et al., 2014; Auer et al., 2019). Sequential allocation and adaptive experimentation have a long statistical history (Robbins, 1952; Lai and Robbins, 1985; Lai, 1987); the expanding set of actions retains that allocation problem and adds an evolving inferential target.

For a fixed arm set, identifying every relatively near-optimal arm is the ALL-ε problem (Mason et al., 2020). Joint reward and identification objectives also have established precedents (Degenne et al., 2019; Yang et al., 2026; Cho et al., 2025). Our task requires continuing reports under arrivals, with all-time label reliability and an explicit cumulative cost for unresolved arm time. The distinction concerns the reporting protocol and its cost, rather than a new arriving-arm model. The simultaneous recovery scales used in our lower bounds build on existing information inequalities and multiple-recovery arguments (Kaufmann et al., 2016; Malloy and Nowak, 2014; Jamieson and Jain, 2018).

We give three results about maintenance cost. First, a common-reference lower bound shows how one uncertain arrival can obstruct many other certifications. Its information queue accumulates cost over time within one reward environment. Second, we obtain matching worst-schedule orders for cumulative unresolved area, including finite-horizon truncation, and derive the polynomial arrival boundary for jointly sublinear area and expected regret. This worst-case lower bound already occurs in an initial static batch; the common-reference result describes a separate consequence of continuing arrivals. Third, on a specified repeated-update family, rapid resolution of temporary

membership tasks has matching inverse-square regret costs. Allowing a constant-factor larger area budget admits a logarithmic regret upper bound. A nonvanishing-probability lower bound holds in one fixed environment under the same cumulative-area requirement. The deferred endpoint has only an upper bound. A separate activation-budgeted witness covers an intermediate interval on a more strongly separated family; the general causal frontier under overlapping arrivals remains open.

Fixed checkpoints make attainable costs explicit, but their quotas can consume a moderate budget before useful reports appear. F-CCP uses every observation to update valid reporting evidence and plans one certificate observation at a time. A predictable priority controls the proposed service rate. Separate budgets limit the virtual-index cost of reward choices and of certification, so exploration by the reward kernel does not exhaust the certification account. The resulting guarantees concern reliability, regret protection, and conditional service bounds; the local planning proxy is not a solution of the full statistical allocation problem. The fixed-checkpoint witnesses are retained for their original guarantees in the supplement.

The components have precedents. Betting processes provide time-uniform inference for bounded means (Waudby-Smith and Ramdas, 2024); our particular mixture and its bounded-reward validity are proved explicitly. Thompson sampling supplies a flexible reward rule (Thompson, 1933). MOTS obtains fixed-arm minimax guarantees by modifying sampled scores (Jin et al., 2021). Our protection instead constrains the virtual-index cost of the actual selected action under arrivals, and does not transfer MOTS's theorem to a cold-temperature Beta rule. Conservative bandits control performance against a fixed baseline (Wu et al., 2016); the two accounts here control index deficits and are not realized shortfalls from a separately run baseline. The spectral accounting and MC-MOSS concentration tools are inputs from the companion work (Temporal Geometry), not a new characterization claimed here.

Information about future task expiration also matters. A schedule-specific comparator can defer a temporary question in a way that one causal policy facing an unknown schedule cannot reproduce uniformly. We retain that information distinction and an activation-budgeted intermediate-regime result alongside the general maintenance bounds. The latter uses its own window conditions and is not an optimality claim for F-CCP.

Section 2 fixes the joint task and separates task parameters from allocation preferences. Section 3 presents F-CCP and its guarantees. Sections 4 and 5 give the general costs, temporary-task bounds, and information limits. Section 6 evaluates frozen configurations, including mean improvements, tail costs, failed reports, and the corrected same-controller comparison.

2 Model, reliable reports, and maintenance cost

2.1 Observation model and regret

Let $\mathcal{A}$ be a possibly infinite set of arms. Arm i arrives at time τ_i, after which it remains available. At every physical time $t = 1, 2, \ldots,$

$$\mathcal{A}_t = \{i : \tau_i \le t\}, \qquad 1 \le K_t = |\mathcal{A}_t| < \infty.$$

Arbitrary finite batches are allowed, including $K_T > T$ at an evaluation endpoint. Each arm has an unknown stationary reward law on $[0,1]$ with mean r_i. Its successive potential observations $(Y_{i,n})_{n\geq 1}$ are independent and identically distributed, and the streams of different arms are independent. Arrivals are exogenous and independent of both these streams and policy randomization. Unless stated otherwise, each result is for a fixed deterministic arrival schedule.

At time t, the policy first receives all current arrivals, then selects one arm $I_t \in \mathcal{A}_t$, observes one reward X_{t,I_t}, and finally issues its report. It uses the observed history

$$H_{t-1} = (\mathcal{A}_s, I_s, X_{s,I_s})_{s=1}^{t-1}$$

and independent randomization. Means, future arrivals, and true membership labels are unavailable to the policy. With $N_i(t) = \sum_{s\leq t} \mathbf{1}\{I_s = i\}$, the physical observation is $X_{t,I_t} = Y_{I_t,N_{I_t}(t-1)+1}$. Write

$$\hat{r}_i(n) = \frac{1}{n}\sum_{s=1}^{n} Y_{i,s}, \qquad r^*(t) = \max_{i\in\mathcal{A}_t} r_i, \qquad \Delta_i(t) = r^*(t) - r_i \quad (i \in \mathcal{A}_t).$$

At a deterministic endpoint T, cumulative dynamic regret is

$$R_T(\pi) = \sum_{t=1}^{T} \Delta_{I_t}(t).$$

This is a sum of mean gaps, random through the policy's arm selections; $\mathbb{E}R_T$ is its expectation. The quantity does not compare realized reward sequences. Physical time always counts actual observations.

2.2 Reliable reports and cumulative unresolved time

For $0 < \varepsilon < 1$, the target set is

$$\mathcal{G}_t^{\varepsilon} = \{i \in \mathcal{A}_t : \Delta_i(t) \leq \varepsilon\}.$$

After the current observation, a report partitions all of $\mathcal{A}_t$ into $\mathcal{P}_t, \mathcal{N}_t, \mathcal{V}_t$: certified members, certified nonmembers, and unresolved arms. Define

$$E_{\text{lab}} = \{\text{for all } t:\ \mathcal{P}_t \subseteq \mathcal{G}_t^{\varepsilon} \subseteq \mathcal{A}_t \setminus \mathcal{N}_t\}.$$

A policy is δ-reliable if $\mathbb{P}(E_{\text{lab}}) \geq 1-\delta$ in every allowed reward environment. A positive report may be withdrawn after a new arrival. On the correctness event, a negative report remains valid because $r^*(t)$ is nondecreasing and the arm's mean is stationary.

Reliability permits all arms to remain unresolved. For a boundary width $0 < \eta < 1$, define

$$\begin{aligned} d_i^{\varepsilon}(t) &= |\Delta_i(t) - \varepsilon|, \\ U_t(\eta) &= \sum_{i\in\mathcal{A}_t} \mathbf{1}\{i \in \mathcal{V}_t, d_i^{\varepsilon}(t) > \eta\}, \qquad \mathcal{U}_T(\eta) = \sum_{t=1}^{T} U_t(\eta). \end{aligned} \tag{1}$$

The count includes newly arrived arms with no observations and old arms that become unresolved again. Every such arm–time pair is charged; the loss is not limited to the time before a first decision. Within the boundary band, unresolved time is exempt, but any label issued there must still be correct. The true distance $d_i^{\varepsilon}(t)$ is used only by the evaluator and in the analysis.

Task parameters are fixed within a comparison. Allocation preferences change the strategy, rather than the reporting target. The parameters used below have the following roles.

Parameter	Role
$\varepsilon, \eta \in (0,1)$	Membership tolerance and exempt boundary width
$\delta \in (0, 1/2)$	Simultaneous reporting error budget
$\lambda_t \in [0,1]$	Predictable preference for certification
$p_0 \in (0,1),\ k_0 > 0,\ \nu > 0$	Service floor, backlog scale, and planning price floor
$\mathsf{B},\ \theta > 0$	Reward-score kernel and its temperature
$\beta_R, \beta_C \in [0, \infty]$	Fixed reward and certification guard coefficients
ρ	Auxiliary checkpoint precision; F-CCP uses $\eta/4$

The fixed-checkpoint witnesses may prescribe other precisions before observations. Preference switching does not change $\varepsilon, \eta, \delta$, create observations, or reset accumulated evidence or budget accounts.

The main requirement at a fixed T is

$$\mathbb{E}_\vartheta R_T \le r, \qquad \mathbb{P}_\vartheta\{E_{\mathrm{lab}}, \mathcal{U}_T(\eta) \le a\} \ge 1 - \delta. \tag{2}$$

It combines an expected regret bound with a joint high-probability label–area guarantee. In particular, $\mathcal{U}_T = o(T)$ requires the time-average number of unresolved separated arms to vanish, rather than only a small unresolved fraction of all arms. The area guarantee is for each deterministic endpoint; it need not hold for every endpoint on one common area event.

For a performance family Θ, let $\mathfrak{R}_\Theta(a)$ be the infimum of $\sup_{\vartheta \in \Theta} \mathbb{E}_\vartheta R_T$ over policies reliable on the full bounded model and satisfying the probability condition in (2) for every $\vartheta \in \Theta$. The value is $+\infty$ if the feasible class is empty. Full-model label reliability and a family-specific area guarantee are distinct requirements. The lower bounds remain valid when reliability is required only on the lower-bound subfamily.

2.3 How arrivals change the reporting problem

For a fixed old arm, $\Delta_i(t)$ is nondecreasing, so its true membership can change only from member to nonmember. Its boundary distance $|\Delta_i(t) - \varepsilon|$ need not be monotone. A new candidate can first create uncertainty about the current best mean, obstructing positive certification of many old arms even when those arms remain truly near-optimal. A sufficiently good new reference can later move the boundary past earlier candidates, making their fine classification unnecessary. Sections 4 and 5 quantify these two effects under the same reporting protocol.

Arms with close means need not be close to the relative boundary $r^*(t) - \varepsilon$. Conversely, a candidate near that boundary can require fine evidence even when its reward gap is appreciable.

The algorithm's observable work set is therefore defined from confidence intervals, not from the evaluator's true gaps or unresolved area.

3 Flexible certificate-cost planning

The reporting task need not receive a fixed fraction of every sampling budget. Evidence obtained while collecting rewards can already certify some arms, and a small amount of additional work can complete other reports. Flexible Certificate-Cost Planning (F-CCP) uses one set of lifetime statistics for both purposes. Its service preference controls how often certification is attempted; two separate budgets protect the reward allocation and reserve capacity for certification. The resulting guarantees concern reliable labels, regret order, and service completion. They do not identify the optimal allocation for every joint regret–area constraint.

3.1 Evidence and current reporting work

Order arms by arrival, with a fixed order within each batch. Write $n_i = N_i(t-1)$ and Σ_i for the actual count and reward sum available before the action at time t. Initially $n_i = \Sigma_i = 0$ and $[\ell_i, u_i] = [0, 1]$. For $\alpha_i = \delta/\{2i(i+1)\}$, invert the beta-mixture process

$$\mathsf{M}_{i,n}(m) = \frac{\mathrm{B}(\Sigma_{i,n} + 1/2, n - \Sigma_{i,n} + 1/2)}{\mathrm{B}(1/2, 1/2) m^{\Sigma_{i,n}} (1-m)^{n-\Sigma_{i,n}}}, \qquad m \in [0, 1], \tag{3}$$

where B is the beta function, distinguished from the optimal mean $r^*(t)$. The interval is the intersection of $\{m : \mathsf{M}_{i,n}(m) \le 1/\alpha_i\}$ over all observed positive prefixes. At the actual count

$$\rho = \eta/4, \qquad q_i = \left\lceil \frac{2}{\rho^2} \log \frac{4i(i+1)}{\delta} \right\rceil, \tag{4}$$

intersect it also with $[\hat{r}_i(q_i) - \rho/2, \hat{r}_i(q_i) + \rho/2] \cap [0, 1]$. Continue the mixture updates after this checkpoint. The construction is valid for bounded rewards, including nonbinary rewards; its supermartingale argument is given in Supplement S2. Sequential betting and mixture confidence sequences are established tools (Waudby-Smith and Ramdas, 2024).

For a nonempty confidence rectangle define

$$g_i^- = \max\{0, \max_{j \ne i} \ell_j - u_i\}, \qquad g_i^+ = \max\{0, \max_{j \ne i} u_j - \ell_i\}. \tag{5}$$

Both bounds are zero when there is only one arm. They are the exact minimum and maximum of the gap over this rectangle, including its self-exclusion constraint. After observing a reward, report

$$\mathcal{P}_t = \{i : g_i^+ \le \varepsilon\}, \quad \mathcal{N}_t = \{i : g_i^- > \varepsilon\}, \quad \mathcal{V}_t = \mathcal{A}_t \setminus (\mathcal{P}_t \cup \mathcal{N}_t). \tag{6}$$

A negative report does not remove an arm from future reward allocation. Among unresolved arms let

$$Z_t = \{i \in \mathcal{V}_t : [g_i^-, g_i^+] \subseteq [\varepsilon - \eta, \varepsilon + \eta]\}, \qquad W_t = \mathcal{V}_t \setminus Z_t. \tag{7}$$

These formulas are also evaluated before the action, after the current arrivals, when choosing that round's service. An arm in Z_t remains reported as unresolved; the rectangle already shows that it incurs no η-unresolved cost. Recompute the sets after every arrival and observation. An empty interval sets a persistent failure flag: all subsequent reports are unresolved and designated service stops. Reward allocation, its guard, and the actual statistics continue.

3.2 A local certificate plan

The plan assigns observations a positive price and asks which current pairwise certificate is cheapest to pursue. Its centers and radii are

$$m_i = \Pi_{[\ell_i,u_i]}\frac{\Sigma_i + 1/2}{n_i + 1}, \quad v_i^{\text{plan}} = \max(m_i - \ell_i, u_i - m_i), \quad \bar{n}_i = \max(1, n_i). \tag{8}$$

Here $\bar{n}_i$ is a planning convention; no fictitious observation enters the confidence interval. For each $i \in W_t$, take as its reference j the arm of largest m_j if $\max_k m_k - m_i > \varepsilon$; otherwise take the largest-upper-bound arm other than i. Resolve reference ties by arrival order. Set

$$h_i = .95\max\{|m_j - m_i - \varepsilon|, \eta - |m_j - m_i - \varepsilon|\}, \quad c_k = \nu + (1 - \lambda_t)\zeta_t(k), \quad a_k = v_k^{\text{plan}}\sqrt{\bar{n}_k}, \tag{9}$$

where $\nu > 0$ and the observable index gap ζ_t is defined below. Solve the two-coordinate proxy problem

$$\min_{x_i \geq \bar{n}_i,\, x_j \geq \bar{n}_j} c_i(x_i - \bar{n}_i) + c_j(x_j - \bar{n}_j) \quad \text{subject to} \quad a_i/\sqrt{x_i} + a_j/\sqrt{x_j} \leq h_i. \tag{10}$$

The interior solution is explicit; boundary and zero-radius cases appear in Supplement S2. Among pairs with an available endpoint, use the pair of smallest proxy cost. Within that pair, select the endpoint maximizing $v_k^{\text{plan}}/(\bar{n}_k c_k)$ among those with $n_k < q_k$ and $x_k > n_k$ (using the actual count). Execute only one pull, then replan. If no pair provides an endpoint, select the widest interval among arms with $n_k < q_k$, with arrival order breaking ties. Call this subroutine Plan_t, and its proposed arm j_t.

The optimization in (10) is exact for its continuous proxy. It holds the centers fixed and approximates radius reduction by an inverse square root. Other competitors can remain after one pair is resolved. Thus its solution is a scheduling rule, not an information complexity or a proof of optimal statistical allocation.

3.3 Virtual indices and two budgets

The reward guard uses a virtual, clipped MC-MOSS index. Its arrival clock and spectral comparison come from the companion study (Temporal Geometry). Initialize base $= 0$. After arrivals, if $K_t > 1$ and either base $= 0$ or $K_t - 1 \geq 2\,\text{base}$, set base $= K_t - 1$, $\sigma = t$, and $d_{\text{clock}} = 2\,\text{base}$. Retain all

observations. With $s = t - \sigma + 1$, define

$$J_i(t) = \begin{cases} 1, & n_i = 0, \\ \min\{1, \hat{r}_i(n_i) + \sqrt{[\log\{s/(d_{\text{clock}} n_i)\}]_+ / n_i}\}, & n_i > 0, \end{cases} \qquad \zeta_t(i) = \max_j J_j(t) - J_i(t). \quad (11)$$

When $K_t = 1$, set $J_i = 1$ and $\zeta_t(i) = 0$ without using an uninitialized clock. The virtual index never forces an observation by itself. In particular, ζ_t is an observable index difference, not realized regret or a difference between two independent runs.

Choose the first initial arm as a counting anchor. Among all other arrived arms, let $k_{\ell,t}$ count current ages $t - \tau_i + 1$ in group 0 (age one), or group $\ell \geq 1$ (ages in $(2^{\ell-1}, 2^\ell]$). The current spectrum and its running maximum are

$$\mathcal{S}_t = \sum_{\ell \geq 0} \min\{2^\ell, \sqrt{k_{\ell,t} 2^\ell}\}, \qquad \overline{\mathcal{S}}_t = \max_{u \leq t} \mathcal{S}_u. \quad (12)$$

The anchor's identity supplies no information about its mean or optimality. Computing (12) uses only arrived arms' current ages, with no endpoint or future-arrival input.

Fix $\beta_R, \beta_C \in [0, \infty]$ before the run. Maintain separate ledgers, initially zero,

$$C_R(t) = \sum_{u \leq t:\, \text{reward}} \zeta_u(I_u), \qquad C_C(t) = \sum_{u \leq t:\, \text{service}} \zeta_u(I_u). \quad (13)$$

Accept a proposed service only if $C_C(t-1) + \zeta_t(j_t) \leq \beta_C \overline{\mathcal{S}}_t$. Otherwise use a reward round and record a denial. On any reward round maximize the selected reward kernel's scores over

$$\mathcal{F}_R(t) = \{i : C_R(t-1) + \zeta_t(i) \leq \beta_R \overline{\mathcal{S}}_t\}. \quad (14)$$

A largest-J_i arm has zero index gap, so this set is nonempty. An infinite budget disables the corresponding check, including when the spectrum is zero; it is implemented as `None`, rather than as budget zero. No ledger is reset at a clock change or preference change.

For example, the default reward scores are one draw per arm,

$$\Xi_i(t) \sim \text{Beta}(1 + \Sigma_i/\theta,\ 1 + (n_i - \Sigma_i)/\theta). \quad (15)$$

For Bernoulli rewards, $\theta = 1$ gives standard Beta$(1,1)$ Thompson sampling (Thompson, 1933); $\theta = .5$ is a colder action rule. For general bounded rewards these fractional-count scores are a sampling heuristic. The evidence always uses the original counts and reward sums. On a reward round draw the full score vector once, then maximize that same vector over (14); do not redraw following rejection. UCB1, Growing-MOSS, MC-MOSS, and KL-UCB can also supply reward scores.

Table 1: F-CCP presets. The default reward kernel is Beta sampling. An inactive service budget is never checked because service is off. The guarantees refer to the conditions stated in this section.

Preset (data ID)	λ/control	θ	β_R	β_C
Reward (F-R)	0/off	.5	16	inactive
Balanced (F-auto-0.6)	.6/adaptive	.5	16	1
Classification (F-area)	1/fixed	.5	16	∞
Compatibility (F-R-TS-exact)	0/off	1	∞	inactive

3.4 Service preference and executable rule

Let $\lambda_t \in [0,1]$ be a predictable service preference. Given a proposed service j_t, write $b_t = |W_t|$ and $\widehat{c}_t = \max_i m_i - m_{j_t}$. The adaptive controller uses

$$w_t = p_0 + (1-p_0)\frac{b_t}{b_t + k_0}\frac{\varepsilon}{\varepsilon + \widehat{c}_t}, \qquad p_t = \begin{cases} 0, & b_t = 0 \text{ or service is disabled,} \\ \dfrac{\lambda_t w_t}{1 - \lambda_t + \lambda_t w_t}, & \text{otherwise.} \end{cases} \tag{16}$$

Fixed-rate control instead uses $p_t = \lambda_t \mathbf{1}\{b_t > 0\}$. Service is disabled when the failure flag is set or $\lambda_t = 0$. The presets use $p_0 = .02$, $k_0 = 4$, and $\nu = .1$. Few or costly proposed certificates tend to reduce the service rate; a large, inexpensive backlog tends to increase it. The controller sees W_t, not the true unresolved area. Neither λ_t nor w_t is an estimated optimal multiplier or a forecast of future arrivals.

Use a phase $\phi_0 = 0$: add p_t each round, and if the result is at least one, attempt one service and subtract one. A denial does not refund this phase. In exact arithmetic the number of attempts through T is $\lfloor \sum_{t \le T} p_t \rfloor$. For fixed $\lambda > 0$, on busy rounds,

$$p_t \ge p_{\min}(\lambda) := \frac{\lambda p_0}{1 - \lambda + \lambda p_0} > 0. \tag{17}$$

There is no positive service lower bound at $\lambda = 0$. Table 1 separates the four presets. Changing only `set_priority(1)` does not disable an existing certification budget. Preference changes occur between completed physical rounds and retain the phase, evidence, counts, and both ledgers.

Within a planned pair, equal endpoint scores favor the candidate i; other ties use the fixed arrival-order scans. The procedure sees no means, hidden identities, future arrivals, or endpoint. Stopping at an arbitrary endpoint leaves the last completed report unchanged. Its straightforward implementation takes $O(K_t^2)$ work per round when pair planning or the self-excluding gap bounds are evaluated by nested scans, and $O(K_t)$ stored arm statistics. Numerical root finding and floating comparisons are described separately in the supplement.

Algorithm 1 Flexible certificate-cost planning (F-CCP)

Require: Fixed $\varepsilon, \eta, \delta$, budgets β_R, β_C, reward kernel, temperature, controller parameters; predictable λ_t.

Initialize empty arm statistics, intervals, and quotas; base $= 0$, $\phi = C_R = C_C = \overline{\mathcal{S}} = 0$; failure $=$ false.
for each physical time $t = 1, 2, \ldots$ **do**
  Admit the current batch in arrival order with $n_i = \Sigma_i = 0$, $[\ell_i, u_i] = [0, 1]$, birth t, and quota (4).
  Update the arrival clock and calculate $J_i(t), \zeta_t(i)$ using (11) (use the singleton convention).
  Update $\overline{\mathcal{S}} \leftarrow \max\{\overline{\mathcal{S}}, \mathcal{S}_t\}$ using current ages.
  Set attempt $=$ false, service $=$ false, $p_t = 0$.
  if not failed and $\lambda_t > 0$ **then**
    Recompute the pre-action reports and W_t from current intervals.
    if $W_t \neq \varnothing$ **then**
      $j_t \leftarrow \mathrm{Plan}_t$ using (8)–(10); calculate p_t.
      $\phi \leftarrow \phi + p_t$.
      if $\phi \geq 1$ **then**
        $\phi \leftarrow \phi - 1$; attempt $=$ true.
      end if
    end if
  end if
  if attempt and a legal j_t exists **then**
    if the service guard is off or $C_C + \zeta_t(j_t) \leq \beta_C \overline{\mathcal{S}}$ **then**
      $I_t \leftarrow j_t$; service $=$ true; $C_C \leftarrow C_C + \zeta_t(I_t)$.
    else
      Record one service denial; retain the decremented phase.
    end if
  end if
  if not service **then**
    Generate the reward score vector once from the current actual history.
    $I_t \leftarrow$ first maximizer of these scores over $\mathcal{F}_R(t)$ (all $\mathcal{A}_t$ when that guard is off).
    $C_R \leftarrow C_R + \zeta_t(I_t)$.
  end if
  Pull I_t once; observe X_{t,I_t}; $n_{I_t} \leftarrow n_{I_t} + 1$, $\Sigma_{I_t} \leftarrow \Sigma_{I_t} + X_{t,I_t}$.
  if not failed **then**
    Intersect the selected arm's interval with its new mixture interval; at $n_{I_t} = q_{I_t}$ also intersect the fixed-prefix interval.
    If any interval is empty, set failure $=$ true permanently.
  end if
  Report (6), or $(\varnothing, \varnothing, \mathcal{A}_t)$ if failed.
end for

3.5 Reliability, reward protection, and service completion

Proposition 3.1 (Reliable reports)**.** *Under the bounded stationary model, every legal predictable choice of preferences, prices, and service actions in F-CCP satisfies* $\mathbb{P}(E_{\mathrm{lab}}) \geq 1-\delta$*. On a common event* $\mathcal{E}$ *of this probability, all confidence intervals cover their means at all observed counts, all reports are correct, and every arm in* Z_t *lies in the boundary band.*

The evidence argument does not require the proxy plan to predict its actual sampling cost. The mixture and checkpoint errors each receive $\delta/2$, and a union bound gives $\mathcal{E}$.

Proposition 3.2 (Compatibility endpoint)**.** *Fix a legal reward kernel* B*. If service is off from the start* ($\lambda_t \equiv 0$)*,* $\beta_R = \infty$*, and the complete kernel state, initialization, and random-number source agree with those of* B*, the two procedures have identical actions, lifetime counts, and regret on every potential reward realization and arrival schedule.*

This is a same-backend pathwise statement. A common seed in different random-number backends does not provide the required coupling. The colder, guarded F-R preset is distinct from this endpoint. Switching off service after using it does not recreate a baseline's independent past.

The regret comparison uses an arbitrary-action inequality,

$$\mathbb{E}R_T \leq \min\left\{T, 300\mathcal{S}_T + \mathbb{E}\sum_{t\leq T}\zeta_t(I_t)\right\}. \tag{18}$$

Supplement S3 proves the bridge from the companion MC-MOSS potential-prefix and arrival-clock bounds to arbitrary actions; the potential-prefix bounds apply to the actual sequence of selected actions.

Theorem 3.3 (Two-budget reward protection)**.** *Fix finite nonnegative* β_R, β_C *and use legal pre-observation actions. For every deterministic schedule and endpoint,*

$$\mathbb{E}R_T \leq \min\{T, (300 + 2\beta_R + 2\beta_C)\mathcal{S}_T\}. \tag{19}$$

If service is off throughout the run, omit $2\beta_C$ *even when its guard is disabled. If* $\beta_R < \infty$ *and* $\beta_C = \infty$*, then, with* $Q_T^{\mathrm{cap}} = \sum_{i\in\mathcal{A}_T} q_i$*,*

$$\mathbb{E}R_T \leq \min\{T, (300 + 2\beta_R)\mathcal{S}_T + Q_T^{\mathrm{cap}}\}. \tag{20}$$

These bounds are unconditional, including reporting failures.

The ledgers satisfy their respective running-spectrum budgets, and $\overline{\mathcal{S}}_T \leq 2\mathcal{S}_T$. Substitution in (18) proves the finite-budget bound. Unlimited certification costs at most Q_T^{cap} actual service pulls. Thus F-R has the conservative bound $332\mathcal{S}_T$, Balanced has $334\mathcal{S}_T$, and Classification retains the extra Q_T^{cap} term. The companion spectral minimax lower bound makes the finite-budget regret order sharp, for fixed coefficients; the spectral characterization itself is an existing input (Temporal Geometry). None of these bounds compares F-CCP's realized regret with an independently run Thompson sampler.

Theorem 3.4 (Area under a positive service rate)**.** *Suppose every pre-action busy round has $p_t \geq p_{\min} > 0$. Let D_T^{srv} be the number of service attempts rejected by the certification budget. On $\mathcal{E}$, simultaneously at every endpoint,*

$$\mathcal{U}_T(\eta) \leq K_T \min\left\{T, \frac{Q_T^{\mathrm{cap}} + D_T^{\mathrm{srv}} + 1}{p_{\min}}\right\}. \tag{21}$$

When $D_T^{\mathrm{srv}} = 0$, this is $O(\min\{K_T T, K_T^2 \eta^{-2} \log(K_T/\delta)/p_{\min}\})$.

Here busy means that the current W_t is nonempty after arrivals and before observation. If every current arm has reached its checkpoint, the gap intervals are narrow enough that W_t is empty. Each successful service therefore advances a finite remaining checkpoint, and phase accounting bounds the number of busy rounds. This proves the theorem without requiring a cost-optimal plan. Classification has $p_{\min} = 1$ and no refusals, hence the worst-area order matches the general lower bound in its common parameter range. Finite-budget Balanced must retain the denial term; it has no unconditional prescribed-area guarantee. An arbitrary predictable λ_t preserves label reliability, but completion speed needs the displayed positive lower bound.

Corollary 3.5 (Finite-arm completion)**.** *Suppose $\beta_C > 0$ is fixed, busy rounds have a uniform positive service rate, and arrivals eventually stop with finitely many arms. On $\mathcal{E}$, every arm outside the boundary band eventually receives a definite label. This conclusion holds regardless of continuing reward-kernel exploration and its separate ledger consumption.*

A conservative finite-prefix bound, the compatibility proof, and a one-round obstruction to exact all-environment dominance appear in Supplement S4. Completion on a fixed-confidence event concerns the boundary band; it is not unconditional almost-sure consistency for the exact oracle set.

4 Fundamental costs of continual maintenance

4.1 One uncertain arrival can obstruct many reports

The first lower bound retains one reward environment throughout the time axis. It therefore measures a cumulative burden without adding the worst cases of incompatible experiments. Fix $0 < \varepsilon \leq 1/16$, $0 < \delta \leq 1/16$, and $0 < \eta < \varepsilon/2$. An initial reference has Bernoulli mean $1/4 + \varepsilon/2$. Candidates arrive on any prescribed schedule. In environment zero each has mean $p = 1/4$; environment j changes only candidate j to mean $p + 2\varepsilon$. All candidates are good in environment zero, with boundary distance $\varepsilon/2$. After candidate j arrives in environment j, every other candidate is bad.

Let M_t count arrived candidates and $a_t = M_t - M_{t-1}$. Define

$$q_- = \left\lfloor \frac{\mathrm{kl}(1/4, \delta)}{\mathrm{kl}(1/4, 1/4 + 2\varepsilon)} \right\rfloor, \qquad Q_t = (Q_{t-1} + q_- a_t - 1)_+, \quad Q_0 = 0,$$
$$\mathcal{B}_T = \sum_{t \leq T} (M_t - 1)_+ \mathbf{1}\{Q_t > 0\}. \tag{22}$$

This analytical queue is an information-capacity certificate: arrivals add q_- units per candidate, while only one observation is possible per round. It is distinct from F-CCP's phased service rule and from the fixed-checkpoint witness's FIFO queue. It imposes no pathwise sampling quota on every candidate. In the stated range $q_- \geq 2$.

Theorem 4.1 (Broadcast uncertainty)**.** *For every policy reliable on environment zero and all its one-candidate alternatives,*

$$\mathbb{E}_0 \mathcal{U}_T(\eta) \geq (3/4 - \delta)\mathcal{B}_T.$$

If $\mathcal{B}_T > 0$, then

$$\mathbb{P}_0\{\mathcal{U}_T(\eta) \geq \mathcal{B}_T/4\} \geq 1/4.$$

These conclusions still hold when the policy knows the full arrival schedule, the horizon, and the reference distribution.

At a busy time, some deterministic candidate label has expected sample count below q_-. Changing that candidate alone has too little history KL to permit positive certification of another candidate with probability at least $1/4$. Reliability also bounds incorrect negative reports. The remaining probability must be unresolved. The same reference environment supports this argument at every time; the difficult label is selected from expected counts rather than from a favorable realized history. The supplementary proof includes the bounded-variable step needed for the probability statement.

If m candidates have arrived by time s, the queue obeys the deterministic inequality

$$\sum_{t \leq s+qm-1} (M_t - 1)_+ \mathbf{1}\{Q_t(q) > 0\} \geq \frac{q-1}{2} m(m-1), \qquad q \geq 2. \tag{23}$$

The factor m^2 reflects the number of affected reports as well as the work needed to exclude high-quality alternatives. The subtraction of one unit in $q - 1$ accounts for the last observation of each busy period: reports are issued after service.

4.2 Worst-schedule maintenance area

Let $\mathfrak{A}_{T,K}$ be the smallest a attainable by a full-model reliable policy with

$$\sup_{\boldsymbol{\tau}: K_T \leq K} \sup_{\boldsymbol{r}} \mathbb{P}_{\boldsymbol{r}}\{E_{\text{lab}}^c \text{ or } \mathcal{U}_T(\eta) > a\} \leq \delta.$$

This optimization imposes no regret budget. Policies may know T, K and the schedule for the lower bound. The upper bound uses neither T nor K as input.

Theorem 4.2 (Worst-schedule area and joint attainability)**.** *For integers $K \geq 2$ and $T \geq 1$, $0 < \varepsilon \leq 1/8$, $0 < \eta \leq \varepsilon/32$, and $0 < \delta \leq 1/16$,*

$$2^{-24} \min\left\{KT, \frac{K^2}{\eta^2} \log \frac{K}{\delta}\right\} \leq \mathfrak{A}_{T,K} \leq 256 \min\left\{KT, \frac{K^2}{\eta^2} \log \frac{K}{\delta}\right\}. \tag{24}$$

Checkpoint-UCB with $\rho = \eta/4$ attains the upper order.

The same Checkpoint-UCB witness also satisfies

$$\mathbb{E}R_T \le \min\left\{T, K + 4\sqrt{3KT\log(2T)} + 1 + 99K\eta^{-2}\log(K/\delta)\right\}. \tag{25}$$

For the upper bound, completed checkpoints imply endpoint errors at most $\eta/4$, sufficient to classify all arms outside the boundary band. The unfinished-work queue is dominated by adding q_i at each arrival and serving at even rounds. On the simultaneous coverage event,

$$\mathcal{U}_T(\eta) \le \min\left\{K_T T, 2K_T \sum_{i \le K_T} q_i\right\}. \tag{26}$$

Additional UCB observations reduce the work; they are not charged again as new checkpoint samples. The lower bound uses $K-1$ candidates with independent means $1/2 \pm 2\eta$ and a reference of mean $1/2 + \varepsilon$, all available at the first pull. Simultaneous partial recovery and a horizon truncation give the probability statement required by the definition of $\mathfrak{A}_{T,K}$. The statistical $\log K$ factor has fixed-arm multiple-testing precedents; this worst-schedule lower construction itself requires no repeated arrivals.

The classification-priority mode of F-CCP, with $\lambda = 1$ and no certification guard, attains the same worst-area order by Theorem 3.4. Its constants and reward bound are stated separately. The explicit constants in (24) and (25) continue to refer to the fixed-checkpoint witness in Supplementary Section S5.

4.3 Growth rates and the limits of total-count criteria

Corollary 4.3 (Arrival-growth boundary)**.** *Fix the task parameters of Theorem 4.1 and one infinite arrival schedule. If a policy is reliable for every reward environment and has a deterministic envelope $a_T = o(T)$ satisfying (2) in every environment, then $K_t = o(\sqrt{t})$. Conversely, $K_T^2 \log(eK_T) = o(T)$ suffices for one horizon-free policy to have both $\mathbb{E}R_T = o(T)$ and such a sublinear area envelope. In particular, for $K_t \asymp t^\alpha$, $0 < \alpha < 1$, simultaneous attainment is possible if and only if $\alpha < 1/2$.*

The necessity follows from (23). For sufficiency, the general checkpoint completion bound in Supplementary Lemma S5.2 applies throughout $0 < \eta < \varepsilon/2$, together with Supplementary Proposition S5.1. This argument does not require the narrower range $\eta \le \varepsilon/32$ in Theorem 4.2. The exponent statement leaves a logarithmic gap for general critical schedules.

Within the parameter range of Theorem 4.2, (24) also gives a uniform terminal-count statement: over all schedules with $K_T \le k(T)$, sublinear expected regret and sublinear service envelopes are jointly attainable if and only if $k(T)^2 \log k(T) = o(T)$. Here the worst schedule can change with T. This statement concerns a different schedule class from the single infinite schedule in Corollary 4.3.

The necessary condition $K_t = o(\sqrt{t})$ is not sufficient on every fixed schedule. For example, batches of size

$$m_n = \left\lfloor \frac{\sqrt{s_n}}{(\log s_n)^{1/4}} \right\rfloor, \qquad s_n = 2^{2^n},$$

give $K_t = o(\sqrt{t})$ but preclude any $o(T)$ area envelope uniform over reward environments. Near s_n, simultaneous classification of that burst requires accumulated unresolved area of order $m_n^2 \log m_n$, which exceeds s_n by a growing factor. The supplementary argument selects a single infinite schedule; the difficult reward environment is allowed to vary with the endpoint, as required by the stated uniform guarantee.

5 Temporary tasks and limits of the regret–maintenance trade-off

5.1 A repeated-update family

We keep the reward environment, arrival schedule, endpoint, and reporting parameters fixed when comparing two area budgets. Choose integers $J \geq 1$, $M \geq \max\{2, J\}$ and

$$0 < \varepsilon \leq 1/8, \quad 0 < \delta \leq 1/16, \quad 0 < v, \quad Jv \leq \varepsilon/4, \quad 0 < d \leq v/32, \quad \eta = d/2. \tag{27}$$

At $a_e = 1 + eH$, $e = 0, \ldots, J$, a reference with mean $b_e = 1/2 + ev$ arrives. For each $e < J$, the same batch contains M candidates with means

$$r_{e,k} = b_e - \varepsilon + \sigma_{e,k} d, \qquad \sigma_{e,k} \in \{-1, +1\}. \tag{28}$$

All rewards are Bernoulli and all arms remain available. The last batch contains only its reference, and $T = (J + 1)H$. The family Θ includes every sign configuration and every within-batch permutation. The policy is not told which arm is the reference. Lower bounds may disclose those identities and the schedule.

In its birth window a candidate is good exactly when its sign is positive. Its reward gap is $\varepsilon - \sigma_{e,k} d$, while its distance to the membership boundary is d. At the next update its gap becomes $\varepsilon + v - \sigma_{e,k} d$, so both signs are bad, with boundary distance at least $v - d$. Every reference remains good: the final good set consists of all references. The tasks that distinguish the candidate signs expire even though their arms remain available. The restriction $Jv \leq \varepsilon/4$ limits the number of updates at any fixed positive improvement size.

5.2 Low area forces enough temporary tasks to finish

The lower-bound ingredient permits arbitrary interleaving and stopping. For m independent uniform signs with observation laws P_i^+, P_i^-, put

$$\kappa = \max_i \max\{D(P_i^+ \| P_i^-), D(P_i^- \| P_i^+)\} \in (0, \infty).$$

If a bounded stopping time has at least half the signs determined with prior probability at least $1/2$, and all configurations have familywise report error at most $\delta \leq 1/16$, then

$$\overline{\mathbb{E}} \sum_{i=1}^{m} N_i \geq \frac{m}{1024\kappa} \log \frac{m}{\delta}. \tag{29}$$

Other observations must not encode these signs. The proof factors the conditional posterior after the adaptive sampling and bounded stopping kernels cancel, and combines simultaneous recovery with coordinatewise change of measure (Kaufmann et al., 2016). It does not assume independence of the random sample counts. The product prior is a proof device; the final lower bound concerns a fixed environment.

A window that never reaches half-completion contributes at least $MH/2$ area. Thus, on the event $\mathcal{U}_T(\eta) \le V/8$, where $V = JMH$, at least $3J/4$ windows half-complete. A common product prior over all windows turns this area requirement into enough instances of (29). Each candidate observation costs at least $\varepsilon - d$ regret, and observations from distinct batches are charged once. This argument does not impose a deadline on every individual batch; the policy can choose where to defer. The heterogeneous completion probabilities and their information-allocation relaxation are given in Supplementary Section S8. That relaxation is a necessary constraint, with no general causal attainability claim.

5.3 Shared evidence and the two endpoint witnesses

A saved upper endpoint for an old candidate can be compared with the lower endpoint of a later reference. In the separated-window construction,

$$u_{e,k} < \ell_{e+1,\mathrm{ref}} - \varepsilon$$

certifies exclusion after the new reference matures. No new observation of that old candidate is needed for this report, although the reward layer remains free to select it. Reference identities in this explanation are analytical roles, not information supplied to the policy. Supplementary Proposition S9.1 gives the complete heterogeneous conditions, unresolved-area bound, and unconditional regret bound for Checkpoint-UCB. Each quota is charged once over the arm's lifetime, and the birth-window condition is essential. These guarantees do not apply to arbitrary F-CCP configurations by renaming the method.

5.4 Rapid and deferred maintenance at the same endpoint

For the homogeneous family (27)–(28), set

$$K = J(M+1)+1, \quad L_K = \log \frac{4K(K+1)}{\delta}, \quad \rho_f = d/4, \quad \rho_c = v/8,$$
$$q_f = \lceil 32 L_K/d^2 \rceil, \quad q_c = \lceil 128 L_K/v^2 \rceil, \quad H = 128(M+J+1)q_f, \quad V = JMH, \quad T = (J+1)H. \tag{30}$$

The quantities q_f, q_c are uniform analysis bounds on the actual armwise quotas. Neither final arm count nor future times are policy inputs. Fine precision can be set from the task input as $\rho_f = \eta/2$; the coarse precision is a separate preselected configuration. Automatic adaptation to an unknown improvement scale is not asserted.

Theorem 5.1 (Rapid and deferred maintenance at the same endpoint)**.** *On the preceding family and horizon,*

$$2^{-20}\frac{\varepsilon JM}{d^2}\log\frac{M}{\delta} \le \mathfrak{R}_\Theta(V/8) \le 2^{18}\frac{\varepsilon JM}{d^2}\log\frac{M}{\delta}. \tag{31}$$

The coarse-precision Checkpoint-UCB policy has joint label-and-area success probability at least $1-\delta$ *with area budget* $9V/8$*, and*

$$\sup_{\vartheta\in\Theta}\mathbb{E}_\vartheta R_T \le 2\varepsilon JMq_c + 64\left(\frac{JM}{\varepsilon}+\frac{J^2}{v}\right)\log(2T)+1. \tag{32}$$

Consequently, at fixed legal $J, M, \varepsilon, v, \delta$ *and* $d \downarrow 0$*, with* $\eta = d/2$ *and the horizon in* (30)*,* $\mathfrak{R}_\Theta(V/8) = \Theta(d^{-2})$*, whereas* $\mathfrak{R}_\Theta(9V/8) = O(\log(1/d))$*.*

For the lower bound, low area forces enough windows to satisfy (29). For the upper bound, (18) gives the two area budgets. At the specified horizon the deterministic bound $R_T \le 2\varepsilon T$ suffices for the rapid order; it would not justify the same conclusion at arbitrarily longer windows. The coarse bound instead counts checkpoint observations and excess ordinary UCB pulls separately, as in (19).

The two budgets concern the same $\Theta, \varepsilon, \eta, \delta, T$, but allow different service levels. Their ratio does not describe an individual delay for every arm. The theorem does not claim that the coarse policy meets the rapid budget, that its regret upper bound is matched, or that all intermediate budgets are characterized on this family and horizon. A preselected mixture of fine and coarse batches gives additional attainable points in the supplement. The separation concerns this specified repeated-update family, with η and T changing together as $d \downarrow 0$.

5.5 A probability lower bound under the same area requirement

An expectation lower bound alone could reflect unusual sample paths. The following result gives a stronger conclusion without replacing cumulative area by individual deadlines.

Theorem 5.2 (A nonvanishing probability of the rapid-maintenance cost)**.** *In the repeated-update family, suppose a reliable policy satisfies*

$$\mathbb{P}_\sigma\{E_{\mathrm{lab}}, \mathcal{U}_T(\eta) \le V/8\} \ge 1-\delta$$

for every sign configuration. Then one fixed configuration satisfies

$$\mathbb{P}_\sigma\left\{R_T \ge 2^{-20}\frac{\varepsilon JM}{d^2}\log\frac{M}{\delta}\right\} \ge \frac{11}{80}. \tag{33}$$

The proof stops each window at half-completion, its endpoint, or its $(q+1)$st candidate observation, where q is a fixed sample budget. Stopping at the first observation beyond q retains reports that become determinate after a reference observation while the candidate count is still q. Applying the partial-recovery lemma to this bounded experiment makes the probability of a cheap completion less than $1/2$. Low total area requires many completions; counting cheap ones then leaves enough expensive windows with probability at least $1/5 - \delta \ge 11/80$. Their actual

observations give (33). No independence between window-completion events is needed. Averaging under the common prior then selects one fixed environment.

Since $11/80 > \delta$, the rapid endpoint in (31) also has matching order when regret and area must be bounded on one event of probability at least $1 - \delta$. The exact joint-quantile definition and corollary are in Supplementary Section S10. This consequence concerns the rapid endpoint at the prescribed horizon and gives no matching joint-quantile result for deferred maintenance.

5.6 Information limits and intermediate service levels

A policy designed separately for each known schedule has more information than one causal policy required to work across an unknown schedule class. Both still lack the true reward means. Supplementary Theorem S12.1 constructs early and late improvements with the same horizon, area budget, final arm count, and common initial history. Timely service on the late schedule forces an inverse-square cost on the early schedule, while an early-schedule comparator has a logarithmic upper bound. This rules out a universal constant-factor comparison with every separately optimized schedule frontier. It does not rule out an optimal policy for a common causal minimax problem.

There is also a positive intermediate-budget result on a more strongly separated repeated-update family. Write $m = JM$, $V = mH$, and $k(a) = \lceil (V - a)/H \rceil$. The activation-budgeted maintenance witness (ABM) uses coarse lifetime checkpoints, then at most s fine candidates and corresponding references. Fine evidence is collected from new observation blocks after predictable activation, with confidence budgets assigned by activation rank. Under the separate window conditions in Supplementary Theorem S13.4, for $A_{\min} \le a < V$,

$$\frac{\varepsilon k(a)}{3072 d^2} \log \frac{k(a)}{\delta} \le \mathfrak{R}_{\Theta_H}(a) \le R_0(T) + 1024 \frac{\varepsilon k(a)}{d^2} \log \frac{k(a)}{\delta}.$$

The supplement defines $A_{\min}$ and the retained additive reward cost $R_0(T)$, and gives the full construction and proof. This family uses a different H from (30). Given s, the witness does not need future arrivals; calibrating s from an area target uses the performance family's m, H and other analysis quantities. The interval result belongs to ABM, not to the preference parameter of F-CCP, and leaves smaller budgets and general overlapping arrivals unresolved.

6 Numerical studies

6.1 Protocol and comparisons

We study reward performance and the cost of reliable reporting separately. The frozen main suite contains 12 scenarios, 24 configurations and 96 paired replications per scenario and configuration, giving 27,648 records. A subsequent correction of one ALL comparator contributes 1,152 additional records. The original and corrected comparator records are both retained. The primary analysis replaces the original adaptive ALL configuration by its corrected version, leaving 24 configurations and 27,648 records.

The scenarios have 2–20 arms and horizons of 12,000–16,000. Eleven use Bernoulli rewards; one

uses bounded two-point rewards. The reporting parameters ε and η vary by scenario, while $\delta = .05$ throughout. Supplementary Section S15 gives every arrival time, mean and configuration. Within a replication, all methods share the same potential reward stream for each arm, and the mean labels of arms arriving together are jointly permuted. The action routines receive only revealed arms, past observations and their own randomization; they do not receive the means, future arrivals or evaluation horizon. All failed-report paths remain in every reported mean, area and quantile.

The five reward comparators are UCB1, Growing-MOSS, MC-MOSS, Thompson sampling and KL-UCB (Garivier and Cappé, 2011). They make no designated certification pulls and carry the same valid reporter as F-CCP. Thompson sampling uses Beta$(1, 1)$ updates for Bernoulli rewards; its fractional updates for the bounded two-point scenario are an action heuristic, not a correctly specified Bernoulli posterior. F-R uses temperature .5 and reward protection $\beta_R = 16$ with service disabled. The compatibility configuration F-R-TS-exact instead uses temperature one and no reward guard; it reproduces the specified Thompson rule rather than serving as a second independent comparator.

For joint performance we compare the prespecified F configurations with CCP-original, ALL-area and ALL-auto-costmatched. CCP-original is the earlier CCP configuration in this test suite, not the checkpoint method in Supplementary Section S17. The ALL methods are natural retained-history adaptations, rather than implementations of the original FAREAST or $(ST)^2$ procedures. The corrected adaptive ALL method uses the same projected-mean sampling price as F-CCP, the same reward kernel and the same two budgets. Its initial implementation returned zero as that price; the correction changes that comparator input alone, with the scenarios, seeds and all F-CCP parameters fixed. The superseded output remains a diagnostic in Supplementary Section S16.

Development and testing use separate environments and seeds. The reward configuration was selected by the prespecified equal-weight development mean regret, and the moderate-service configuration was fixed before the main test. No preference, temperature or confidence parameter is selected from the test results. The parameter values in these experiments are not generally in the narrower regime of the minimax lower bound; these comparisons do not validate its asymptotic rate.

6.2 Reward performance with service disabled

F-R has a lower mean regret than all five reward comparators in 11 of the 12 scenarios, and a strictly lower mean in 59 of the 60 scenario–comparator comparisons. Its mean is 19.5% below Thompson sampling when the 12 scenario means are averaged with equal weights. This ratio compares two descriptive suite averages; it is not an average of scenario-specific percentage reductions or a risk under one common environment distribution. The gain in this experiment concerns the reward-score configuration and its protection, since the certificate scheduler is disabled.

The two-close-arms scenario is an exception. F-R minus Thompson has mean 0.70 and a paired, two-sided 95% approximate t interval $[-5.28, 6.68]$. Across the full family of 60 prespecified comparisons, 52 one-sided Bonferroni-adjusted upper endpoints are below zero. Under the separate, more permissive margin of 5% of the comparator mean plus two regret units, 54 are below the corresponding threshold. The latter criterion is not zero-margin noninferiority. Neither the low

Table 2: Mean cumulative dynamic regret in the frozen test suite. Each entry uses 96 paired replications; Monte Carlo standard errors are in Supplementary Table 3. All six methods devote no pulls to certification.

Scenario	UCB1	G-MOSS	MC-MOSS	TS	KL-UCB	F-R
Two close arms	75.52	52.16	52.16	51.37	62.63	52.07
Few arrivals	32.60	10.71	11.02	7.47	12.52	5.63
Stable benchmark	393.14	98.11	92.20	66.47	181.01	41.90
Successive improvements	313.99	105.44	86.29	57.61	123.14	53.59
Burst	532.82	140.79	131.95	80.01	177.09	68.56
Membership boundary	482.45	115.74	113.09	83.32	225.66	53.57
Many close means	341.20	188.60	190.01	202.27	267.46	169.80
Late improvement	364.87	126.45	101.93	83.89	196.49	70.56
Rare rewards	146.50	102.35	81.43	60.16	78.06	50.29
Ceiling	366.96	126.06	102.01	28.42	59.24	26.46
Larger pool	702.32	189.70	179.02	121.25	254.50	98.74
Bounded low variance	365.81	111.25	88.96	62.89	171.08	37.61

Table 3: Descriptive equal-weight averages of the 12 scenario means. The first six rows keep $\beta_C = 1$ wherever certification is active; F-area uses no certification cap. Area is in 10^3 arm-time units. These operating points are not an optimal frontier.

Method / preference	β_C	Regret	Area / 10^3
F-R	inactive	60.73	62.22
F, $\lambda = 0.1$	1	114.68	48.29
F, $\lambda = 0.3$	1	213.55	38.46
F, $\lambda = 0.6$	1	392.99	31.70
F, $\lambda = 0.9$	1	425.03	30.84
F, $\lambda = 1$	1	421.52	31.00
F-area	∞	1210.02	27.12
CCP-original	1	440.32	30.07
ALL-auto-costmatched	1	435.68	36.72
ALL-area	∞	1581.72	36.81

means nor a failure to detect a difference proves uniform dominance over the baselines.

6.3 Preference and certification cost

Increasing service preference reduces unresolved arm-time over much of the tested range, at a substantial reward cost. The points at $\lambda = .9$ and 1 under the fixed certification budget are not monotone: the latter has slightly lower regret and slightly larger area. We retain that reversal. F-area removes the certification cap; it is a different budget configuration and is not a continuation of the same-budget curve.

Relative to ALL-auto-costmatched, F-auto-0.6 has lower equal-weight mean regret (392.99 versus 435.68) and area (31.70 versus 36.72 thousand arm-time units). The comparison with CCP-original is a trade-off: F-auto-0.6 has lower mean regret but higher mean area. F-area has lower suite mean regret and area than ALL-area. These are finite-suite averages, rather than per-scenario or pathwise Pareto guarantees. Supplementary Tables 4 and 6 give the scenario decomposition and

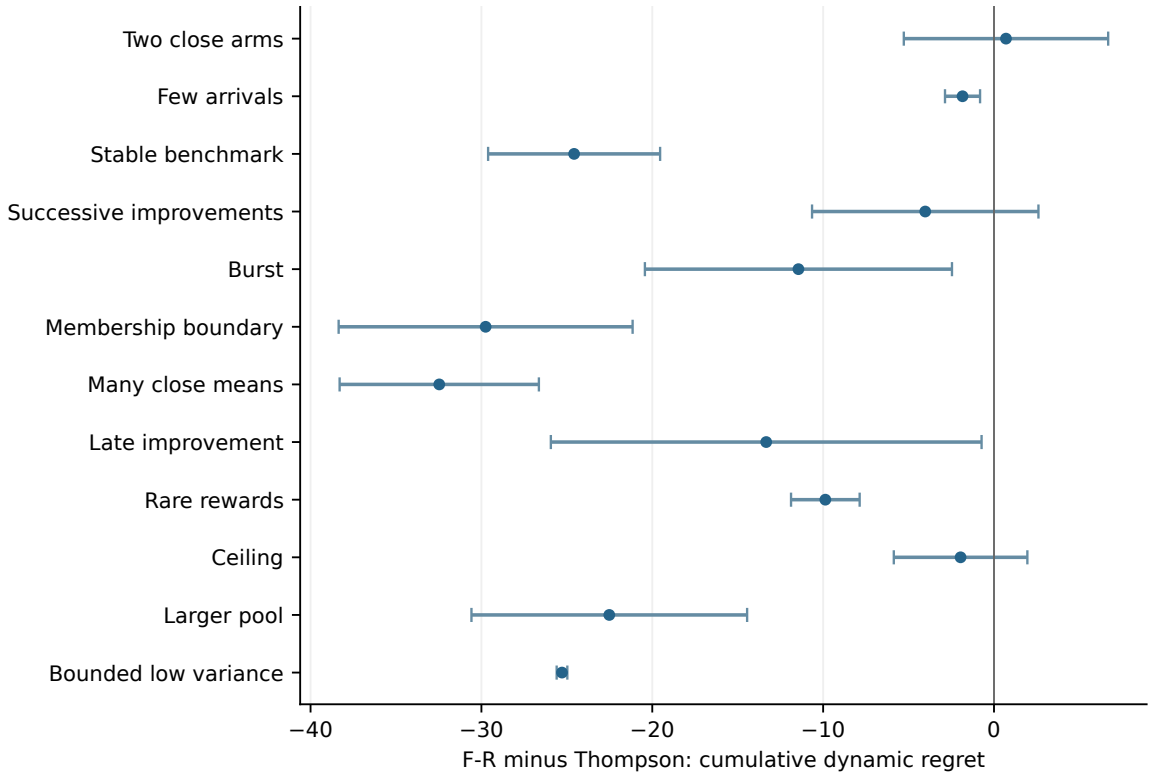


Figure 1: F-R minus Thompson cumulative dynamic regret in each frozen scenario. Points are means of 96 within-replication differences; error bars are individual two-sided 95% approximate t intervals. Negative differences favor F-R. These 12 intervals do not by themselves control the 60-comparison family; the adjusted counts are reported in the text.

paired uncertainty.

The controller's proposed rate differs from the realized service fraction because the phase must accumulate and the certification guard can reject a proposal. Supplementary Table 7 reports proposed busy-round rates, successful pulls, denials and the final unresolved count. Fixed-rate, uncapped-service, temperature-one and unprotected-reward variants are retained as prespecified ablations. We use those comparisons only for the mechanisms they actually change.

6.4 Tail behavior and report failures

The lower mean regret of F-R does not extend to all tail summaries. Its empirical 95th percentile exceeds Thompson's in the burst, ceiling and larger-pool scenarios in Table 4. Each percentile uses only 96 replications; it should not be read as a precise population tail estimate.

F-auto-0.6 produces an incorrect determined label on one of its 1,152 runs; F-R and F-area produce none. Coverage failures and empty intersections also occur: the respective counts are $(6, 1)$ for F-R, $(7, 1)$ for F-auto-0.6, and $(8, 0)$ for F-area. An empty intersection triggers persistent all-unresolved reporting, not successful completion. All these runs enter the cost calculations. Since methods share potential streams, their failures are dependent and cannot be pooled as independent

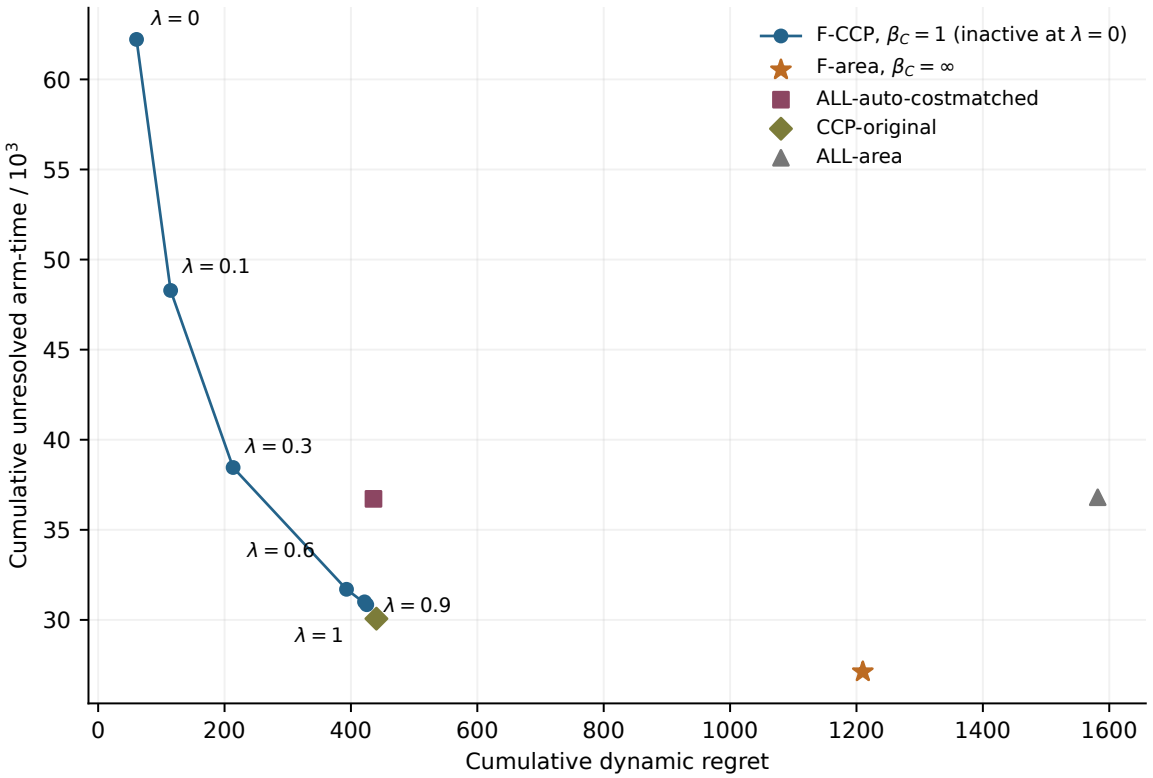


Figure 2: Descriptive averages of terminal costs over the 12 frozen scenarios. Connected circles follow the prespecified order $\lambda = 0, .1, .3, .6, .9, 1$, with $\beta_R = 16$, temperature .5, and $\beta_C = 1$ whenever service is active. The star is F-area with $\beta_C = \infty$; the other markers are named comparators. The connecting segments identify configurations and do not estimate a continuous or optimal frontier. Area is divided by 10^3.

reliability trials. These empirical counts neither replace the reliability theorem nor justify an all-times claim from 96 repetitions. Full failure counts, empirical maxima and implementation qualifications appear in Supplementary Sections S15–S16.

7 Discussion

Reliable maintenance has costs that reward allocation alone does not measure. An uncertain arrival can obstruct many positive reports, while a later improvement can end a fine membership task. The general lower bounds quantify these effects. The worst-area lower bound answers a different question and is already attained by an initial static batch; the fixed-schedule and terminal-count growth consequences retain different quantifiers.

F-CCP adds an implementable choice of emphasis to this task. Its evidence layer uses every observation, its certificate planner proposes a single next observation, and separate accounts control reward and certification costs. Finite guards protect the spectral regret order but can reject service; the area bound retains that rejection cost. Classification priority with no certification guard

Table 4: Empirical 95th percentiles of cumulative dynamic regret from 96 replications per method and scenario. These are finite-sample tail summaries, not high-probability regret bounds.

Scenario	F-R	Thompson
Burst	208.887	123.069
Ceiling	90.940	48.971
Larger pool	237.301	184.175

achieves the worst-area order, without being fastest on every instance. A compatibility endpoint recovers a specified baseline from the start under identical state and randomization. The default cold-temperature reward mode has a different contract.

The frozen experiments support lower mean reward costs for F-R in most of the specified comparisons and useful finite-budget trade-offs for the service modes. They also show an exception in the two-arm close-mean case, heavier empirical regret tails than Thompson sampling in three environments, and a nonzero label-error record for the balanced mode. Pure-reward improvements cannot be attributed to certificate scheduling that was switched off. The corrected matched-cost ALL comparison is distinct from the retained superseded controller result. These findings concern the frozen suite, not uniform dominance or a general frontier.

The activation-budgeted witness characterizes an intermediate interval on a sufficiently separated family, with an additive reward term and its own information requirements. General causal frontiers under unknown and overlapping arrivals remain open. The schedule-information separation does not make every causal comparison impossible. Likewise, the one-step zero-remainder counterexample leaves room for comparisons with an explicit adaptation cost or a restricted performance family.

Automatic precision, reward kernels with better tail behavior, a faithful adaptation of the original ALL-ε algorithms, and large-pool planning costs need further study. The supplement's eventual exact-set consistency theorem belongs to a separate revisable multiresolution procedure under finite final arrivals and strict boundary separation. F-CCP's coverage-event completion outside a boundary band is a different statement.

Data and code availability

The accompanying materials contain simulated data, frozen configurations, code, and the scripts used to generate the article's tables and figures. The principal test has 27,648 recorded runs; a subsequent 1,152-run matched-cost ALL correction is retained separately, together with the superseded comparator. Replacing that comparator for the principal analysis leaves 24 configurations and 27,648 records. Earlier checkpoint comparisons contain 5,760 records and are reported separately in the supplement; the preceding CCP and frontier research suites retain their own protocols and archives. All reported failures remain in the analysis. The available implementation distinguishes the frozen dual-account production engine from the interactive interface and the development branches. These materials are provided locally with the project; no public repository or permanent archive identifier has been verified for this version.

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